

Module-3 : MATHEMATICAL TOOLS OF QUANTUM MECHANICS

Section A

1. Define a linear vector space and list any three properties that its vectors must satisfy.
2. What is a Hilbert space? Mention the role of the scalar product in defining a Hilbert space.
3. Define the scalar product of two state vectors $|\psi\rangle$ and $|\phi\rangle$.
4. State the conjugate symmetry property of the scalar product.
5. Write the bra corresponding to the ket $|\psi\rangle = a|\Phi_1\rangle + b|\Phi_2\rangle$
6. Define the Hermitian adjoint $\hat{A}^\dagger$ of a linear operator $\hat{A}$ in terms of inner products. State the condition under which an operator is strictly Hermitian.
7. Using basic commutation relation $[x, p] = i\hbar$ evaluate $[x, p^2]$ and $[x^2, p]$
8. What is meant by a function $f(\hat{A})$ of an operator?
9. If $\hat{A}|\alpha\rangle = \alpha|\alpha\rangle$, what is $f(\hat{A})|\alpha\rangle$?
10. What can be said about simultaneous eigenvectors of commuting operators?
11. What is a degenerate eigenvalue?
12. What is the position-space wave function of a state $|\psi\rangle$?
13. What is the momentum-space wave function of a state $|\psi\rangle$?
14. How are position and momentum wave functions connected?
15. What mathematical transformation connects the position and momentum representations?

Section B

1. Explain why the set of square-integrable functions is suitable for representing wave functions.
2. Show that the function $\psi(x) = A e^{-a|x|}$, $a > 0$, is square-integrable. Find the normalization constant A .
3. Prove the Schwarz inequality: $|\langle \phi | \psi \rangle|^2 \leq \langle \phi | \phi \rangle \langle \psi | \psi \rangle$
4. Show that if $\hat{A}$ and $\hat{B}$ are Hermitian, then $\hat{A} \hat{B}$ is Hermitian only if $\hat{A}$ and $\hat{B}$ commute.
5. Derive the general uncertainty relation $\Delta \hat{A} \Delta \hat{B} \geq \frac{1}{2} |\langle [\hat{A}, \hat{B}] \rangle|$.
6. Prove that the eigenvalues of a Hermitian operator are real.
7. Prove that eigenvectors belonging to distinct eigenvalues of a Hermitian operator are orthogonal.
8. Write the matrix representation of the ket $|\psi\rangle = a|1\rangle + b|2\rangle$, in the basis $\{|1\rangle, |2\rangle\}$.
9. Find the matrix representation of the operator $\hat{A} = 2|1\rangle\langle 1| + 3|2\rangle\langle 2|$.
10. Show how an operator transforms under a change of basis.
11. Explain how the position-space and momentum-space wave functions are related by a Fourier transform. Write the integral transformation from $\psi(x)$ to $\phi(p)$. Explain the physical meaning of $|\psi(x)|^2$ and $|\phi(p)|^2$.
12. In a two-dimensional Hilbert space spanned by the orthonormal basis $\{|1\rangle, |2\rangle\}$

an operator $\hat{A}$ acts on the basis vectors as follows:

$$\hat{A}|1\rangle = 2|1\rangle + i|2\rangle \quad \text{and} \quad \hat{A}|2\rangle = -i|1\rangle + 3|2\rangle$$

- Construct the 2×2 matrix representation of $\hat{A}$.
 - Determine whether $\hat{A}$ is Hermitian.
 - Find the eigenvalues of $\hat{A}$.
13. Given a state vector $|\psi\rangle$, its position-space wavefunction is $\psi(x) = \langle x|\psi\rangle$ and its momentum-space wavefunction is $\phi(p) = \langle p|\psi\rangle$. Using the continuous completeness relation, show that $\psi(x)$ and $\phi(p)$ form a Fourier transform pair.

Section C

- Hermitian operators represent physical observables in quantum mechanics. Formulate and prove the three major theorems concerning Hermitian operators:
 - Theorem 1: The eigenvalues of a Hermitian operator are strictly real.
 - Theorem 2: The eigenvectors of a Hermitian operator corresponding to distinct eigenvalues are mutually orthogonal.
 - Theorem 3: The eigenvectors of a Hermitian operator form a complete orthonormal basis for the vector space.
- Starting from the Schwarz inequality for two state vectors $|\alpha\rangle = (\hat{A} - \langle\hat{A}\rangle)|\psi\rangle$ and $|\beta\rangle = (\hat{B} - \langle\hat{B}\rangle)|\psi\rangle$, derive the generalized Robertson-Schrödinger uncertainty relation for any two Hermitian operators $\hat{A}$ and $\hat{B}$: $\Delta\hat{A}\Delta\hat{B} \geq \frac{1}{2}|\langle[\hat{A}, \hat{B}]\rangle|$
 - Apply this general relation to position $\hat{x}$ and momentum $\hat{p}_x$ to obtain the classic Heisenberg Uncertainty Principle $\Delta\hat{x}\Delta\hat{p}_x \geq \frac{\hbar}{2}$
- Explain how a state vector $|\psi\rangle$ is represented as a column matrix and an operator $\hat{A}$ is represented as a square matrix in a discrete orthonormal basis $\{|\phi_n\rangle\}$.

- Formulate the matrix eigenvalue problem $\hat{A}|\psi\rangle = \lambda|\psi\rangle$ as a secular equation $\det(A - \lambda I) = 0$
- Demonstrate how a unitary transformation $\hat{U}$ diagonalizes a Hermitian matrix representation.
- Prove that the trace and eigenvalues of an operator are invariant under unitary transformations.

4.

- Define the continuous position basis states $\{|x\rangle\}$ and momentum basis states $\{|p\rangle\}$, stating their delta-function normalization and completeness relations.
- By solving the differential momentum eigenvalue equation $\hat{p} = \vec{p}|\vec{p}\rangle$ in position representation, derive the scalar product (transformation function):
- $\langle x|p\rangle = \sqrt{\frac{1}{2\pi\hbar}} e^{\frac{i}{\hbar}px}$
- Using this result, prove that the position operator acts as $\hat{x}$ and the momentum operator acts as $-i\hbar \frac{\partial}{\partial x}$ in position space.